

FitGuide: An AI Fitness and Nutrition Coach Agent for Personalized Wellness Planning

Option A — Single AI Agent

Group – 4

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I. Problem Statement

Maintaining a healthy lifestyle is a common goal for many individuals, but achieving it consistently can be challenging. Many people want to lose weight, gain muscle, or simply improve their overall health, yet they often struggle to create structured fitness and nutrition plans that fit their personal circumstances. Beginners especially may not know what exercises to perform, how often to train, what foods support their goals, or how to stay consistent over time.

Although there is a large amount of fitness and nutrition information available online, much of it is generic, overwhelming, or not tailored to individual needs. Users are often forced to piece together information from multiple sources, which can lead to confusion, inconsistency, and lack of motivation. In addition, many existing solutions such as personal trainers or nutrition coaches can be expensive and not accessible to everyone.

To address this problem, we propose FitGuide, an AI-powered fitness and nutrition coaching agent that provides personalized and practical recommendations based on user input. The agent will accept user information such as fitness goals, workout schedule, dietary preferences, available equipment, and progress data. Based on this information, the system will generate customized workout suggestions, meal guidance, calorie and protein targets, and ongoing recommendations to help users stay on track.

The target users of this system include college students, busy professionals, and beginner to intermediate gym-goers who want structured guidance without needing a human coach. This problem is well suited for an AI agent because it requires continuous interaction, personalization, decision-making, and integration of multiple data sources rather than a single static prediction.

This aligns with the course objective of designing an intelligent AI agent system that can understand input, reason over it, use tools, and produce meaningful outputs, rather than simply building a standalone model.

II. Chosen Option

We are choosing Option A: Single AI Agent for this project.

A single-agent architecture is appropriate because the problem can be effectively handled by one intelligent system that performs input understanding, reasoning, and action generation. The agent will receive user data, interpret it using a deep learning model, decide which tools to use, and generate a personalized response.

While a multi-agent system could be designed for this domain, it would introduce additional complexity that is not necessary for the core functionality of this project. The fitness and nutrition planning task does not require multiple specialized agents handing work to each other; instead, a single well-designed agent can handle the workflow efficiently.

Choosing Option A allows us to focus on building a clear, functional, and explainable system, which aligns well with the project requirements and makes it easier to implement, debug, and present.

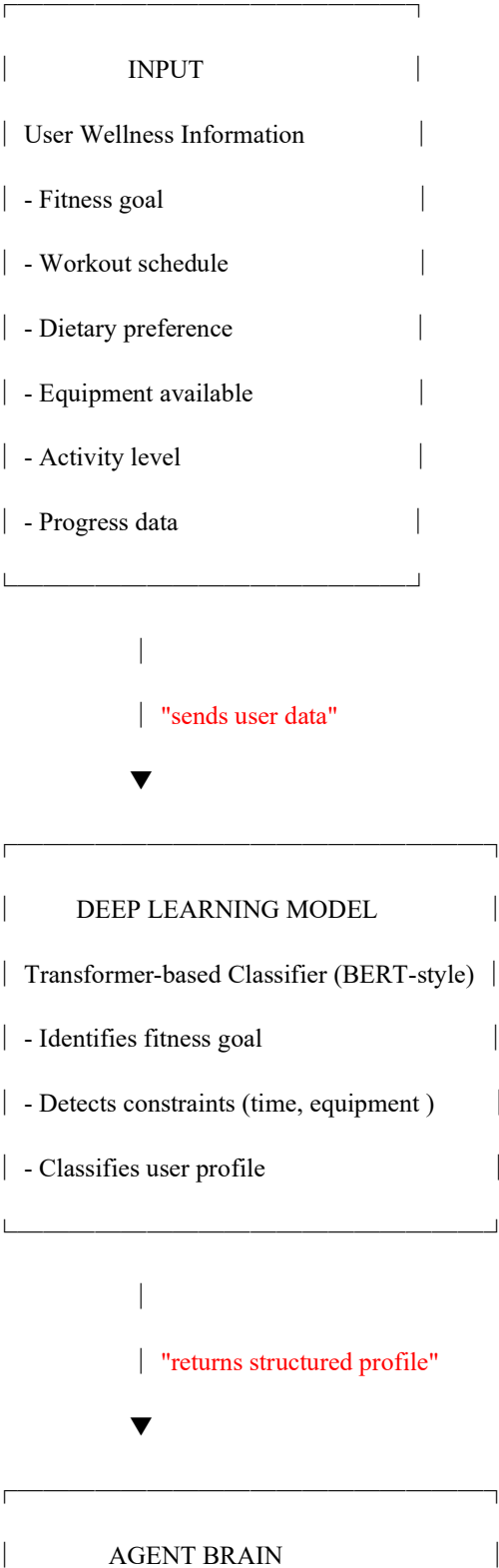
III. Agent Architecture and Diagram

The FitGuide system is designed as a structured AI agent that follows a clear flow of information:

Input → Deep Learning Model → Agent Brain (LLM) → Tools → Output

This architecture reflects the requirement that the deep learning model and the LLM must be separate components, where the deep learning model performs a specific task and the LLM handles reasoning and decision-making.

Architecture Diagram



LLM (Gemini / GPT / Claude)
- Interprets user profile
- Decides which tools to use
- Generates personalized plan

|
| "calls tools & retrieves data"



TOOLS
- Nutrition Database (USDA)
- Exercise Library
- Calorie / Macro Calculator
- Progress Tracker

|
| "returns relevant data"



OUTPUT
Personalized Fitness & Nutrition Plan
- Workout routine
- Meal guidance
- Calorie/protein targets
- Weekly recommendations

Architecture Explanation

The system begins with user input, where the user provides relevant personal and fitness-related information. This data is then processed by a Transformer-based deep learning model, which classifies the user's intent and profile into meaningful categories such as weight loss, muscle gain, or general wellness.

The output of the deep learning model is passed to the LLM agent brain, which acts as the reasoning engine. The LLM uses this structured understanding to decide what actions to take, including selecting the appropriate tools and generating a personalized response.

The tool layer provides structured and reliable data that enhances the system's accuracy. Instead of relying solely on generated text, the agent retrieves information from external sources such as nutrition databases and exercise libraries.

Finally, the system produces an output that is both personalized and actionable. This architecture demonstrates the full AI agent pipeline of perception, reasoning, and action.

IV. Deep Learning Connection

This project integrates multiple deep learning concepts from the course, particularly Transformers and RNN/LSTM-based sequence modeling.

Transformer Models

The first key component is a Transformer-based model, such as BERT or a similar architecture. This model is used to interpret and classify user input written in natural language.

Transformer models are well suited for this task because they can understand context and relationships within sentences. For example, if a user says, "I want to lose fat but only have dumbbells and can train three times a week," the model can extract multiple pieces of information simultaneously.

This allows the system to move beyond keyword matching and instead perform context-aware understanding, which is critical for personalization.

RNN / LSTM Concepts

The second deep learning concept is RNN/LSTM-based sequence modeling, which relates to tracking user progress over time. Fitness and nutrition behavior naturally form sequences, such as daily calorie intake, weekly workout completion, and weight changes.

RNNs and LSTMs are designed to analyze sequential data and identify patterns over time. In this project, these concepts support the idea of adapting recommendations based on user history, even if a full LSTM model is not implemented in the prototype.

Why These Models Fit

These models complement each other:

- Transformers handle understanding user input
- RNN/LSTM concepts support tracking and adapting over time

Together, they provide both personalization and adaptability, which are essential features of the FitGuide system.

V. Agent Framework

We plan to use LangChain as the primary agent framework for this project.

LangChain is well suited for building AI agents that require:

- LLM reasoning
- Tool integration
- Structured workflows
- Modular design

The FitGuide system needs to accept user input, process it, call external tools, and generate a personalized output. LangChain provides built-in support for this type of pipeline, making it easier to connect different components of the system.

We considered other frameworks such as CrewAI and AutoGen, but those are more appropriate for multi-agent systems. Since our project is a single-agent design, LangChain provides a simpler and more direct approach.

Using an established agent framework also satisfies the project requirement that the final implementation must be built using a real agent framework.

VI. Tools and Data

The FitGuide system will use a combination of tools and structured data sources to generate accurate and useful recommendations.

Tools

1. Nutrition Database Tool

This tool retrieves nutritional information such as calories, protein, carbohydrates, and fat for different foods.

2. Exercise Library Tool

This tool stores exercise-related data including muscle groups, required equipment, and recommended sets and reps.

3. Macro / Calorie Calculator

This tool estimates daily calorie and protein needs based on user goals and activity level.

4. Progress Tracker

This tool stores user progress data such as completed workouts and weight trends.

Data Sources

Nutrition Data:

USDA FoodData Central will be used as a reliable source of nutritional information.

Exercise Data:

A custom dataset will be created containing structured exercise information.

Health Guidelines:

Public recommendations from organizations such as the CDC or WHO may be used.

User Data:

User-provided information such as goals, preferences, and progress logs.

Using tools and structured data ensures that the system produces grounded and reliable recommendations instead of relying only on generated text.

VII. Build Plan

The project will be completed over the remaining weeks before the final deadline.

Week 1:

Set up environment, choose LLM, prepare datasets.

Week 2:

Build tools and data layer (nutrition, exercise, calculator).

Week 3:

Implement agent logic using LangChain.

Week 4:

Add progress tracking and improve personalization.

Week 5:

Test system, prepare demo, finalize submission.

This plan is designed to be realistic and focused, ensuring that a working prototype can be completed on time.

VIII. Expected Challenges

1. Generic Recommendations

The system may produce responses that are too general. This will be addressed by using structured data and improving input classification.

2. Architecture Separation

Ensuring a clear distinction between the deep learning model and the LLM may be challenging but will be handled through explicit design.

3. Time Constraints

The project must remain within a manageable scope to meet deadlines.

4. Safety Concerns

The system will be positioned as a general wellness assistant, not a medical authority.

Conclusion

FitGuide is a practical and well-structured AI agent designed to provide personalized fitness and nutrition guidance. It integrates deep learning, reasoning, and tool usage into a system that can help users make better health decisions.

The project aligns with course requirements by demonstrating a clear agent architecture, meaningful use of deep learning models, and a realistic implementation plan. It is both technically appropriate and achievable within the given timeframe, making it a strong candidate for the final project.